The influence of physical process representations on regional and global-scale climate model output

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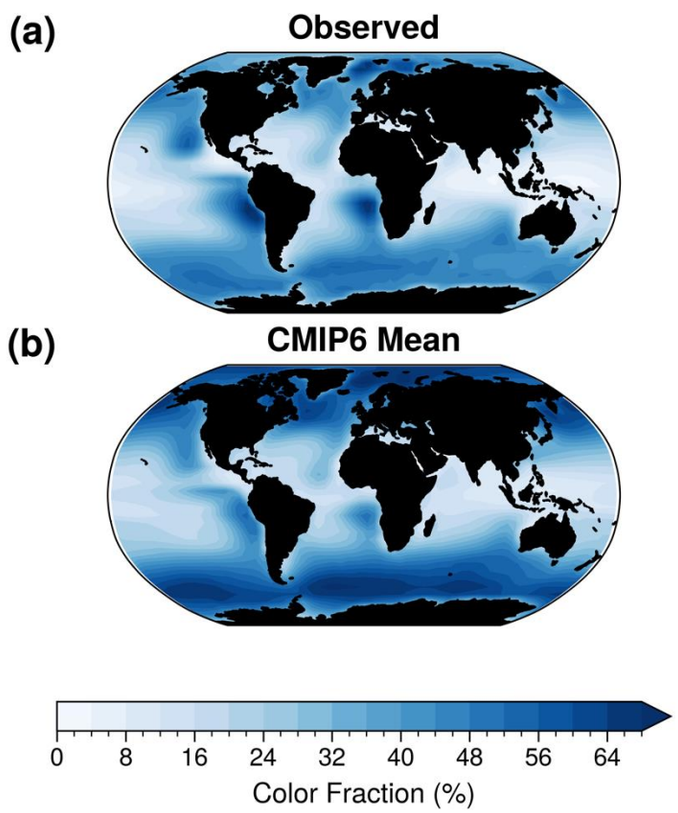
Uncertainty in stratocumulus regions



Low clouds are the **primary source of uncertainty** in climate projections, particularly oceanic tropical low clouds. Climate models **can't accurately capture fine-scale cloud processes** due to computational limitations (Klein et al., 2017).

Where does the uncertainty come from?

Parametric Uncertainty

Uncertainty in model outputs that arises from imprecise or unknown parameter values, assuming the model structure remains fixed

⬇

Reduced by matching the outputs of many model variants to historical observations

vs

Structural Uncertainty

Uncertainty that stems from the design of the model itself (structure, equations, resolution, and simplifications) rather than from parameter or input values

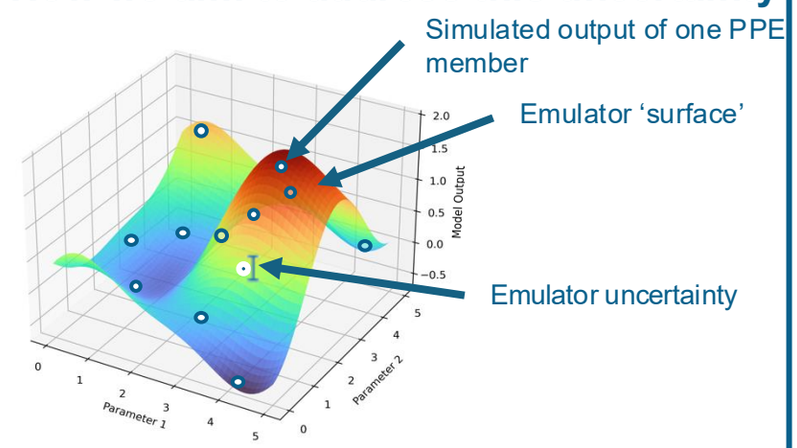
⬇

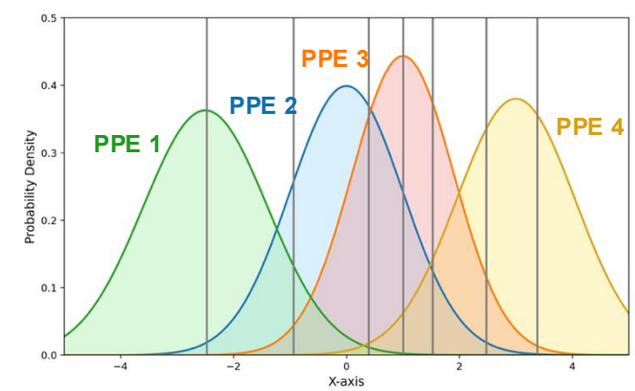
Reduced by changing the model

Model outputs **are influenced** by **both parameter and structural uncertainties**. Therefore, **disentangling these sources of uncertainty** is essential for understanding model behaviour and **improving reliability** (Regayre et al., 2023).

Using **Perturbed Parameter Ensembles (PPEs)**, we can detect **model structural differences** and deficiencies and **define priorities for model development** or simplification (Carslaw et al., 2025).

How we aim to address this uncertainty





PPEs involve running a climate model multiple times with different parameter combinations. By using **Gaussian Process (GP)** emulators, we can expand the sample of input/output relationships, allowing us to have an emulator surface. A more comprehensive approach would be to move towards a **multi-model perturbed parameter ensemble (MM-PPE)**, which would allow for a more thorough and **statistically robust analysis of model uncertainty**.

Regayre, L.A., Deaconu, L., Grosvenor, D.P., Sexton, D.M.H., Symonds, C., Langton, T., Watson-Paris, D., Mulcahy, J.P., Pringle, K.J., Richardson, M., Johnson, J.S., Rostron, J.W., Gordon, H., Lister, G., Stier, P. and Carslaw, K.S. 2023. Identifying climate model structural inconsistencies allows for tight constraint of aerosol radiative forcing. *Atmospheric Chemistry and Physics*. **23**(15), pp.8749–8768.

Carslaw, K.S., Regayre, L.A., Proske, U., Gettelman, A., Sexton, D.M.H., Qian, Y., Marshall, L., Wild, O., Van Lier-Walqui, M., Oertel, A., Peatier, S., Yang, B., Johnson, J.S., Li, S., McCoy, D.T., Sanderson, B.M., Williamson, C.J., Elsaesser, G.S., Yamazaki, K. and Booth, B.B.B. 2025. Opinion: The importance and future development of perturbed parameter ensembles in climate and atmospheric science.

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